

ecosystem at about 700 individuals. The relationship of this estimate to the effective population size (N_e) depends on several factors. Usually, only a few dominant males breed, and it may be difficult for them to locate females, since individuals inhabit such large areas. Moreover, females may reproduce only when there is abundant food. As a result, N_e is only about 25% of the total population size, or about 175 bears.

Because small populations tend to lose genetic variation over time, researchers have used proteins, mitochondrial DNA, and short tandem repeats (see Concept 21.4) to assess genetic variability in the Yellowstone grizzly bear population. All results to date indicate that the Yellowstone population has less genetic variability than other grizzly bear populations in North America.

How might conservation biologists increase the effective size and genetic variation of the Yellowstone grizzly bear population? Migration between isolated populations of grizzlies could increase both effective and total population sizes. Computer models predict that introducing only two unrelated bears each decade into a population of 100 individuals would reduce the loss of genetic variation by about half. For the grizzly bear, and probably for many other species with small populations, finding ways to promote dispersal among populations may be one of the most urgent conservation needs.

This case study and that of the greater prairie chicken show how an understanding of the ecological and evolutionary consequences of small populations can have practical applications in conservation. Next, we look at an alternative approach to understanding the biology of extinction.

Critical Habitat

A loss of critical habitat can cause threatened and endangered populations to show a downward trend, even if the population is far above its minimum viable population size. A focus on the importance of critical habitat emphasizes the environmental factors that can cause a population decline in the first place. If, for instance, an area is deforested, then species that depend on trees will decline in abundance, whether or not they retain genetic variation. The following case study illustrates how a critical-habitat approach has been applied to the conservation of an endangered species.

Case Study: Decline of the Red-Cockaded Woodpecker

The red-cockaded woodpecker (*Picoides borealis*) is found only in the southeastern United States. It requires mature pine forests, preferably ones dominated by the longleaf pine, for its habitat. A critical-habitat factor for the red-cockaded woodpecker is that the undergrowth of plants around the pine trunks must be low (**Figure 56.14a**). Breeding birds tend to abandon nests when vegetation among the pines is thick and higher than about 4.5 m (**Figure 56.14b**); the birds appear to need a clear flight path between their home trees and the neighboring feeding grounds. Periodic fires have

▼ **Figure 56.14** A habitat requirement of the red-cockaded woodpecker.



(a) Forests that can sustain red-cockaded woodpeckers have low undergrowth.



(b) Forests that cannot sustain red-cockaded woodpeckers have high, dense undergrowth that interferes with the woodpeckers' access to feeding grounds.

? How is habitat disturbance absolutely necessary for the long-term survival of the woodpecker?

historically swept through longleaf pine forests, keeping the undergrowth low.

In addition, although most woodpeckers nest in dead trees, the red-cockaded woodpecker drills its nest holes in mature, living pine trees. It also drills small holes around the entrance to its nest cavity, which causes resin from the tree to ooze down the trunk. The resin seems to repel predators, such as corn snakes, that eat bird eggs and nestlings.

One factor leading to decline of the red-cockaded woodpecker has been the destruction or fragmentation of suitable habitats by logging and agriculture (**Figure 56.15**). By recognizing key habitat factors, protecting some longleaf